ASYMPTOTIC SUCCESS UPPER BOUNDS FOR RESTRICTED CRT-LOCAL FACTORING CONTROLLERS AND ECM DUAL-CHAIN COUNTERFAMILIES

ARGUS OPEN RESEARCH PROJECT

Abstract. We prove two unconditional negative theorems against restricted deterministic integer-factoring controller models: (1) for one-gcd CRT-local controllers equipped with equality and Jacobi-symbol feedback, an infinite balanced-semiprime family limits asymptotic success to at most $5/8 + o(1)$; and (2) for the elliptic-curve dual-chain exponent model with resource parameters (c, e), a shell-uniform Rankin and prime-selection construction produces an infinite balanced-semiprime counterfamily with exponent parameter $\eta = 1/(256 \max(3c + 2, e + 1))$.

1. Main Theorems

Theorem 1.1. *Let $N = pq$ range over balanced semiprimes.*

(1) *Every deterministic controller in the frozen one-gcd CRT-local grammar with equality and Jacobi-symbol branching succeeds on a fraction at most $5/8 + o(1)$ of balanced semiprimes as $N \to \infty$.*

(2) *For every fixed positive rational resource tuple (c, e) in the input-specific ECM dual-chain model, there exists an infinite family of balanced semiprimes $N = pq$ on which the dual-chain point multiples return to the identity O simultaneously across both prime factors without separating p and q.*

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